

Four colors suffice for all prime distances on the real line

CONJECTURE 00000000024 | FALSE

Submitted by SucRunBug - AI-assisted mathematical work and formalization

Conjecture

The real-line graph joining points at prime integer distance has countably infinite chromatic number; the chromatic numbers of its finite initial prime-distance graphs tend to infinity.

Proof

Color each real x by $\text{floor}(x)$ modulo 4. If $y=x+p$ for an integer prime p , $\text{floor}(y)=\text{floor}(x)+p$. No prime is divisible by 4, so the two colors differ. Symmetry handles $x=y+p$. This applies to all real points, including negative and nonintegral points.

Consequences

The full graph and every restriction of its prime-distance set admit the same four-coloring. This refutes both claims. The points 0, 2, 5, 7 form a four-clique: their six distances are 2, 5, 7, 3, 5, 2. Hence the full chromatic number is exactly 4, as is that of every initial prime-distance graph containing 7.

Formalization scope

Lean constructs a proper Fin 4-coloring of the actual simple graph on all real numbers and of every restricted prime-distance graph. This proves the upper bound required for the refutation. The supplementary sharpness argument appears in the written proof.

Reproduction

Pinned Lean and Mathlib: v4.33.0. In lean4/: lake exe cache get; lake build; lake env lean Check.lean. The independent reproduce.py checks finite instances; the written general proof and Lean theorems settle the infinite statements.

Source: repository conjectures/00000000024.md; main branch df87a69ed4814eeb34ad161879b53db10c44fdc6.